

Flight Controller Board HARDWARE ARCHITECTURE

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1 Introduction

1.1 Purpose and Scope

This document defines the hardware architecture, component selection rationale, power distribution network (PDN) design, signal integrity (SI) rules, RF subsystem design, electromagnetic compatibility (EMC) strategy, mechanical/DFM constraints, and verification plan for the **HRO-FC1** — a custom flight controller (FC) intended for a multirotor aerial observation platform requiring simultaneous 4K on-board video recording and low-latency analog FPV downlink with remote camera tilt control.

This specification is the authoritative reference for PCB layout, component procurement, and verification test planning. It supersedes the informal concept notes captured in Rev. A.

1.2 Applicable Documents and Standards

- IPC-2221 — Generic Standard on Printed Board Design
- IPC-2141A — Design Guide for High-Speed Controlled Impedance Circuit Boards
- IPC-6012 — Qualification and Performance Specification for Rigid PCBs (Class 2 target, Class 3 workmanship where noted)
- IPC-A-610 — Acceptability of Electronic Assemblies
- FCC Part 15, Subpart C (§15.247 / §15.249) — Intentional radiators, 5.725–5.875 GHz ISM band
- ETSI EN 300 440 — Short-range devices, 5.8 GHz band (EU applicability)
- MIL-STD-810H, Method 514.8 — Vibration (Category 24, rotary-wing platform, as guidance only; full military qualification is out of scope)
- RTCA DO-160 (referenced for environmental test methodology guidance only; full DO-160 qualification not required for this program)

1.3 Definitions and Acronyms

Term	Definition
FC	Flight Controller
PDN	Power Distribution Network
PDS	Power Distribution System (synonymous with PDN in this document)
SI	Signal Integrity
CPWG	Coplanar Waveguide with Ground
VTx	Video Transmitter (analog, 5.8 GHz)
PA	Power Amplifier
IMU	Inertial Measurement Unit
CVBS	Composite Video Baseband Signal (analog composite video)
DFM/DFA	Design for Manufacturing / Design for Assembly
FMEA	Failure Modes and Effects Analysis
EIRP	Effective Isotropic Radiated Power
ESR/ESL	Equivalent Series Resistance / Equivalent Series Inductance

1.4 Mission Profile / Concept of Operations

The vehicle is a hexacopter observation platform performing loiter/hover observation at 30 m to 120 m AGL for missions up to 25 minutes on 6S LiPo. The system must simultaneously: (a) record 4K @ 30 fps to onboard storage for post-mission analysis, (b) provide a real-time analog FPV downlink to the ground pilot/observer for situational awareness, and (c) allow the observer to remotely pitch the camera between -90° (nadir) and $+30^\circ$ from level. Altitude hold via barometer

and GPS position hold are required to keep the platform stable while the observer manually operates the gimbal, decoupling flight control workload from camera aiming workload.

2 System Requirements

Requirements are numbered for traceability into the Verification Matrix (Section 13). “Shall” denotes a mandatory, testable requirement; “should” denotes a goal.

2.1 Functional Requirements

ID	Requirement
REQ-CAM-001	The system shall record video at 4 K (3840×2160) at 30 fps to removable Micro SD storage, U3/V30 rated, up to 128 GB.
REQ-CAM-002	The system shall provide a simultaneous real-time analog CVBS output (NTSC or PAL, field-selectable) for the FPV downlink, independent of the SD recording pipeline.
REQ-CAM-003	The FC shall be able to start/stop SD card recording via a UART command link to the camera module, triggered by a radio control (RC) auxiliary channel.
REQ-GIM-001	The FC shall generate a servo-compatible PWM signal (50 Hz nominal, 1000 μ s to 2000 μ s pulse width) on a hardware timer channel to actuate a camera tilt servo across a minimum mechanical range of 120°.
REQ-GIM-002	Loss of RC signal shall command the gimbal to a pre-configured safe position (default: level, 0°) within 500 ms of failsafe detection.
REQ-VTX-001	The onboard VTx shall be tunable in firmware across the standard 5.8 GHz analog FPV band plan (Section 9) via SPI, with the active channel persisted in non-volatile memory.
REQ-VTX-002	The VTx output power shall be configurable to at least three discrete levels (nominal targets: 25 mW, 200 mW, 400 mW) to support regional regulatory compliance.
REQ-NAV-001	The FC shall maintain barometric altitude hold to within ± 0.5 m in still air and provide GPS-based position hold to within ± 1.5 m horizontal (with ≥ 8 satellites, HDOP < 1.5).
REQ-NAV-002	Compass (magnetometer) heading accuracy shall be within $\pm 5^\circ$ post-calibration, with the module physically separated from high-current conductors per Section 10.

2.2 Performance Requirements

ID	Requirement
REQ-PERF-001	Glass-to-glass analog FPV latency shall not exceed 80 ms (camera sensor to ground-station display), excluding goggle/receiver processing.
REQ-PERF-002	IMU sampling rate shall be ≥ 6.4 kHz raw, with attitude control loop closure at ≥ 2 kHz to support stable hover in gusts up to 8 m/s.
REQ-PERF-003	VTx analog video downlink range (line-of-sight, 400 mW, standard patch/omni antenna) shall achieve ≥ 2 km at ≤ -85 dBm receiver sensitivity margin of 6 dB.

ID	Requirement
REQ-PERF-004	Total continuous current draw of the FC/camera/VTx/gimbal subsystem (excluding motors/ESCs) shall not exceed 3.0 A at 5 V, per the power budget in Table 11.

2.3 Environmental Requirements

ID	Requirement
REQ-ENV-001	Operating temperature range: -10°C to $+45^{\circ}\text{C}$ ambient; storage -20°C to $+60^{\circ}\text{C}$.
REQ-ENV-002	The PCB assembly shall withstand random vibration per MIL-STD-810H Method 514.8, Category 24 (rotary-wing, general), $7.7\text{ g}_{\text{rms}}$, 10 Hz to 2000 Hz, without component detachment or solder joint fatigue over a 2-hour per-axis test.
REQ-ENV-003	The assembly shall tolerate humidity up to 95% RH non-condensing; conformal coating is required per Section 11.
REQ-ENV-004	Operating altitude up to 4000 m MSL (derate convective cooling assumptions accordingly, see Section 6.3).

2.4 EMC / Regulatory Requirements

ID	Requirement
REQ-EMC-001	The VTx intentional radiator shall comply with FCC Part 15.247 (or region-appropriate equivalent, e.g., ETSI EN 300 440) for the target market at the configured output power level.
REQ-EMC-002	Radiated/conducted emissions from the RF section shall not desensitize the IMU, barometer, GPS L1 front-end (1575.42 MHz), or the analog CVBS trace by more than 1 dB / 1 LSB equivalent noise floor increase, verified per Section 10.
REQ-EMC-003	Regulatory caution: 400 mW EIRP exceeds unlicensed limits in several jurisdictions (e.g., EU 25 mW under ETSI EN 300 440 for non-amateur use). The design shall default to the lowest legal power level in firmware and require explicit user configuration to enable higher power modes, with an on-screen/telemetry regulatory disclaimer.

2.5 Safety Requirements

ID	Requirement
REQ-SAF-001	Input power path shall include reverse-polarity protection, input fusing/current limiting, and inrush current limiting for hot-plug of a charged 6S pack (Section 6).
REQ-SAF-002	The MCU shall monitor input bus voltage and enter a low-power/land-now telemetry state below 19.8 V (3.3 V/cell) to protect the LiPo pack.
REQ-SAF-003	Thermal protection: firmware shall throttle or disable the VTx PA if the PA case temperature (via NTC or PA internal flag) exceeds 95°C .

3 System Architecture Overview

3.1 High-Level Block Diagram

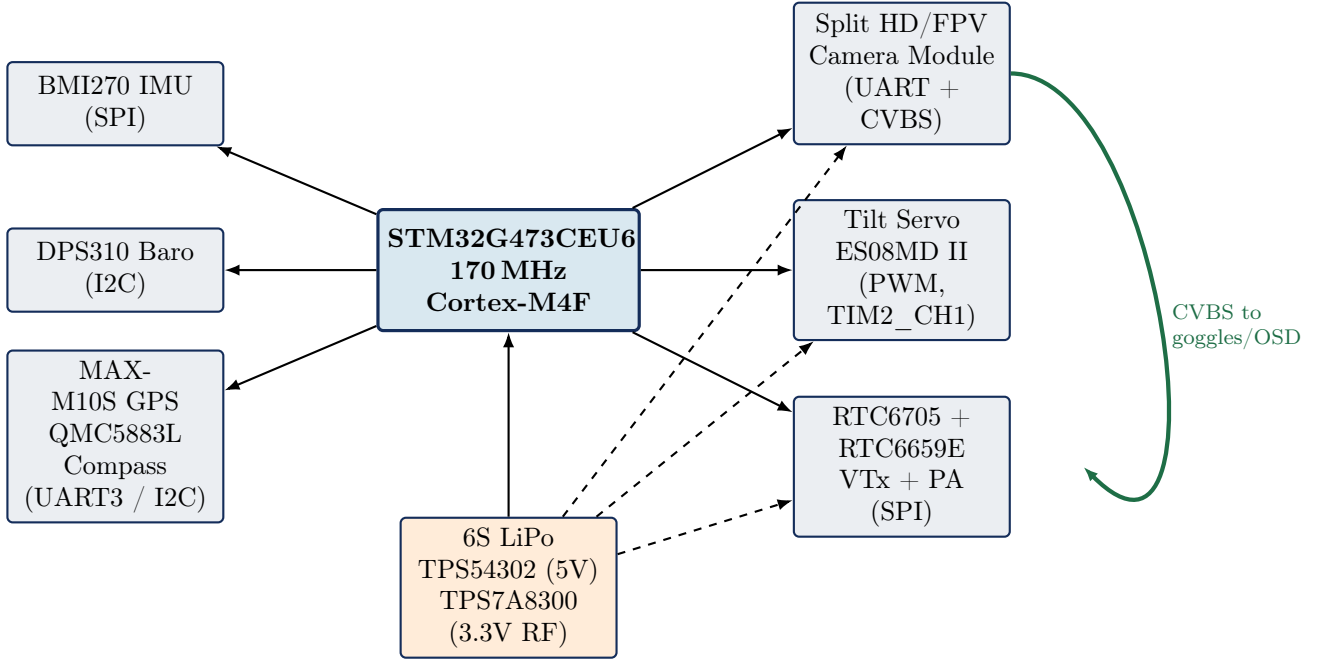


Figure 1: HRO-FC1 top-level hardware block diagram. Solid arrows: control/data signals. Dashed arrows: power rails. Green: analog video signal path, isolated from the digital control path per Section 10.

3.2 Interface Control Summary

4 Camera & Gimbal Subsystem Design

4.1 Camera Selection Rationale

A split-style HD/FPV camera decouples 4K ISP/encode workload from the flight-critical STM32, which only needs to arbitrate a UART control link and, optionally, overlay OSD data onto the CVBS path. This avoids adding an application-processor-class SoC (and its associated Linux/RTOS complexity, boot time, and thermal load) to a safety-relevant flight control board.

Design Note — SD Card Selection: *Micro SD cards shall be U3/V30 (or A2 where the camera ISP supports it) rated to sustain the ≥ 30 MB/s minimum write throughput required for stable 4K@30 H.264/H.265 encoding without frame drops. Industrial-temperature-rated cards (rated to at least the -10°C to $+45^{\circ}\text{C}$ operating range of REQ-ENV-001) are recommended over consumer-grade cards, which are frequently only rated to 0°C at the low end.*

4.2 Gimbal Torque Budget

The tilt servo must overcome the gravitational moment of the camera/lens assembly about the tilt axis plus dynamic (vibration/maneuvering) loads. A first-order sizing calculation:

Assumed camera + mount mass, $m = 25\text{ g} = 0.025\text{ kg}$

Moment arm (CG offset from tilt axis), $r = 20\text{ mm} = 0.02\text{ m}$

Static gravitational torque, $\tau_g = m \cdot g \cdot r = 0.025 \times 9.81 \times 0.02 \approx 4.9 \times 10^{-3}\text{ N} \cdot \text{m} \approx 0.05\text{ kg} \cdot \text{cm}$

Interface	Peripheral	Electrical	Notes
IMU	BMI270	SPI1, 3.3 V, up to 10 MHz	Dedicated CS, no bus sharing with RTC6705
Barometer	DPS310	I2C1, 3.3 V, 400 kHz	Shares I2C bus with compass (different addresses)
GPS	MAX-M10S	UART3, 3.3 V, 115200 bps (config)	PPS input to MCU timer capture for time sync
Compass	QMC5883L	I2C1, 3.3 V, 400 kHz	Located ≥ 25 mm from PA/inductors (Section 10)
Camera control	RunCam Split 4 V3	UART2, 3.3 V TTL, 115200 bps	RunCam Device Protocol (start/stop rec, OSD, mode)
Camera video	RunCam Split 4 V3	CVBS, 1 V _{pp} into 75 Ω	Routed as controlled-impedance analog trace to VTx and to an external OSD/goggle pigtail
Gimbal servo	EMAX ES08MD II	PWM, TIM2_CH1, 3.3 V logic / 5 V supply	Update rate 50 Hz, resolution ≥ 10 -bit equivalent
VTx control	RTC6705	SPI2, 3.3 V, ≤ 1 MHz	Register-based PLL frequency + power level set
Battery sense	—	Analog, resistor-divided to ≤ 3.3 V	ADC1, hardware low-pass at 10 Hz

Table 7: Interface Control Document (ICD) summary. Full pin mapping in Appendix A.

Adding a dynamic safety factor of $6\times$ (to cover angular acceleration during rapid tilt commands, in-flight vibration coupling, and worst-case CG offset with an accessory such as an IR filter or larger lens) gives a required stall torque of approximately $0.3 \text{ kg} \cdot \text{cm}$. The specified EMAX ES08MD II (rated $\sim 2.0 \text{ kg cm}$ at 4.8 V) provides a design margin in excess of $6\times$ over this budget, which comfortably covers gust-induced disturbance torques not captured in the first-order model above.

Design Note — Servo Refresh Rate vs. Control Loop: *The gimbal PWM update (50 Hz) is intentionally decoupled from the 2 kHz+ attitude control loop (REQ-PERF-002). The MCU shall write the target pulse width into a double-buffered compare register (TIM2_CH1 CCR, using the timer’s preload feature) so that updates take effect only at the next PWM period boundary, preventing partial-pulse glitches that could cause servo jitter.*

4.3 Edge Cases and Fault Handling

Candidate	4K Recording	Analog Latency	Notes
RunCam Split 4 V3 (baseline)	4K@30, 2.7K@60	~40 ms typical	Mature UART control protocol (RunCam Device Protocol); wide community/firmware support.
Caddx Loris 4K	4K@30, 4K@60 (HFR)	~45–50 ms	Higher headline resolution/frame-rate options; verify UART control command set parity before substitution.
Foxeer Mix 2	4K@30	~40 ms	Similar form factor; alternate second source.

Table 8: Camera module trade study. RunCam Split 4 V3 selected as primary; Caddx Loris 4K retained as approved alternate pending UART command-set validation.

Edge Case	Required Handling
SD card full	Camera module shall report status via UART telemetry (RunCam Device Protocol status field); FC shall raise an OSD/telemetry warning at $\geq 90\%$ capacity and on write failure. Recording of flight-critical telemetry (if overlaid) must not depend on SD card health.
SD card write corruption on power loss	A minimum 470 μF (bulk, Section 6) plus internal camera-module hold-up capacitance must sustain the camera's 3.3 V/5 V rail for ≥ 50 ms after main bus brown-out to allow the camera's internal file system to close the current video segment/finalize the container (MP4 moov atom). This shall be validated per test VER-PWR-03.
Camera module brown-out / no ACK on UART	FC firmware shall implement a 3-retry UART handshake with 200 ms timeout before flagging a CAM_FAULT status; VTx CVBS path is unaffected since it is a separate analog output not gated by UART link health.
Servo mechanical stall (ice, physical obstruction, wiring snag)	FC shall monitor servo supply current via a low-side shunt on the dedicated servo rail; sustained current > 1.2 A for > 200 ms shall trigger a GIMBAL_STALL flag and command PWM to the neutral position to avoid burning out the servo winding.
RC link loss during active tilt command	Per REQ-GIM-002, revert to configured safe position within 500 ms; do not freeze at last commanded angle, since an extreme nadir/zenith angle could obstruct antenna line-of-sight to the VTx patch antenna.
Sustained recording throttle	4K@30 thermal Camera modules are known to throttle bitrate/frame rate after ~ 10 – 15 minutes of continuous 4K recording in enclosed housings. Provide a minimum 3 mm ventilation gap or thermal pad from the camera processing board to the airframe shell; this is a mechanical integration note for the vehicle housing design, tracked as open item OPEN-01 (Section 15).

5 Bill of Materials & Component Selection

Subsystem	Part Number	Mfr.	Package	Role / Selection Rationale
MCU	STM32G473CEU6	STMicro.	UFQFPN48	170 MHz Cortex-M4F, FPU + CORDIC + FMAC, 6× 16-bit advanced timers for PWM/servo/ESC generation, hardware CRC for SD/telemetry integrity checks.
IMU	BMI270	Bosch	LGA-14	Best-in-class vibration rejection (on-chip anti-aliasing, low-power high-performance modes); SPI up to 10 MHz.
Barometer	DPS310	Infineon	LGA-8	High resolution (± 0.06 hPa RMS), low temperature coefficient, I2C/SPI selectable.
GPS	MAX-M10S	u-blox	LGA-18	Multi-constellation (GPS/GLONASS/Galileo/BeiDou), integrated 10 Hz nav solution, dedicated PPS output.
Compass	QMC5883L	QST	LGA-16	Low-cost 3-axis magnetometer; must be physically separated from ESC/PA current loops – see Section 10.
VTx core	RTC6705	RichWave	QFN-32	Integrated 5.8 GHz FM modulator/PLL synthesizer, SPI-programmable channel and power.
RF PA	RTC6659E	RichWave	QFN-16	Boosts RTC6705 output to ~ 400 mW EIRP class; requires dedicated thermal via array (Section 11).
Main buck	TPS54302DDCR	TI	SOT23-thin/SON-8	Steps 6S (19.8–25.2 V) to 5 V/3 A for camera, servo, digital rail source.
RF LDO	TPS7A8300ARGR	TI	VQFN-20	Ultra-low-noise ($9 \mu\text{V}_{\text{rms}}$ typ.), high PSRR LDO for RTC6705/PA analog supply.
Digital LDO	(TBD, TLV75533)	e.g., TI (proposed)	SOT23-5	Dedicated 3.3 V rail for MCU/IMU/baro/GPS, isolated from RF and servo transients; open selection , see OPEN-02.
Gimbal servo	ES08MD II	EMAX	—	9 g metal-gear digital servo, ~ 2.0 kg cm stall torque at 4.8 V (see Section 4.2 torque budget).
Camera module	RunCam Split 4 V3	RunCam	—	Split lens/processing architecture; 4K@30 SD recording + CVBS output + UART control (Section 4.1).

Table 10: Bill of Materials – key active components. Passive component BOM (decoupling, filtering) is enumerated in Table 12. Full BOM with distributor part numbers, unit cost, and lead time is maintained in the companion procurement spreadsheet (out of scope for this document).

△ Caution — Second-Sourcing: *The RTC6705/RTC6659E pairing and the TPS7A8300 are specified as single-source in this revision. Given global passive/RF-IC supply volatility, a second-source RF chipset (e.g., IRC Tramp-compatible modules) should be evaluated before committing to tooling; this is tracked as OPEN-03.*

6 Power Architecture & Power Integrity (PI)

6.1 Power Budget

6.2 Power Tree

6.3 Buck Converter Design (TPS54302, 5 V/3 A)

Assume a representative switching frequency $f_{sw} = 570$ kHz (typical of this converter class) and a target inductor ripple current of 30% of full load ($\Delta I_L = 0.9$ A):

$$L = \frac{(V_{in,max} - V_{out}) \cdot V_{out}}{V_{in,max} \cdot f_{sw} \cdot \Delta I_L} = \frac{(25.2 - 5.0) \times 5.0}{25.2 \times 570,000 \times 0.9} \approx 7.8 \mu\text{H}$$

A standard 10 μH , shielded-drum inductor rated ≥ 4.5 A_{sat} is selected to provide margin. Duty cycle at nominal/extreme input:

$$D_{nom} = \frac{5.0}{22.2} \approx 22.5\%, \quad D_{min,Vin} = \frac{5.0}{25.2} \approx 19.8\%, \quad D_{max,Vin} = \frac{5.0}{19.8} \approx 25.3\%$$

Load	Rail	Typ. I	Max. I	Typ. P	Max. P
STM32G473 (core+IO)	3.3 V _{DIG}	80 mA	150 mA	0.26 W	0.50 W
BMI270 + DPS310	3.3 V _{DIG}	5 mA	10 mA	0.02 W	0.03 W
MAX-M10S GPS	3.3 V _{DIG}	25 mA	40 mA	0.08 W	0.13 W
QMC5883L	3.3 V _{DIG}	1 mA	2 mA	<0.01 W	0.01 W
Camera module	5 V	400 mA	800 mA	2.0 W	4.0 W
Gimbal servo (ES08MD II)	5 V	150 mA	1500 mA (stall)	0.75 W	7.5 W (transient)
RTC6705 core	3.3 V _{RF}	150 mA	200 mA	0.50 W	0.66 W
RTC6659E PA (400 mW mode)	5 V _{RF}	350 mA	500 mA	1.75 W	2.5 W
Total (5 V rail)	5 V	900 mA	2.8 A	4.5 W	14.0 W (transient)
Total (3.3 V_{DIG})	3.3 V	111 mA	202 mA	0.37 W	0.67 W

Table 11: System power budget. The 3 A continuous rating of the TPS54302 buck (REQ-PERF-004) provides margin over the 0.9 A typical/2.8 A transient 5 V load; servo stall and PA keying transients are handled by bulk capacitance (Section 6.4), not by continuous buck current capability.

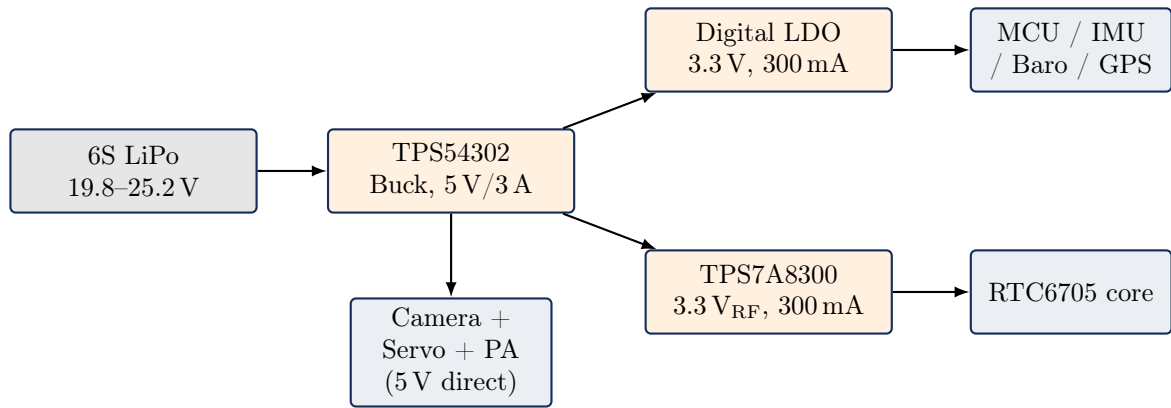


Figure 2: Power tree. Three isolated regulation domains from a single buck pre-regulator: (1) 5 V direct for high-current/noisy loads (camera, servo, PA), (2) 3.3 V_{DIG} for MCU/sensor logic via a quiet digital LDO, (3) 3.3 V_{RF} via ultra-low-noise TPS7A8300 exclusively for the RTC6705 synthesizer core.

Output ripple voltage budget of 50 mV implies a maximum combined capacitor ESR:

$$ESR_{max} = \frac{\Delta V_{ripple}}{\Delta I_L} = \frac{0.05}{0.9} \approx 55 \text{ m}\Omega$$

This is satisfied by a parallel combination of ceramic (bulk high-frequency, low ESL) and polymer/electrolytic (bulk energy storage) capacitance – see Section 6.4.

Design Note — Efficiency and Thermal Derating: At $I_{out} = 2.8 \text{ A}$ transient and $V_{in} = 22.2 \text{ V}$, converter losses (conduction + switching, estimated at $\sim 8\text{--}10\%$ for this converter class at this duty cycle) yield an estimated dissipation of 1.0–1.4 W in the regulator package. With no forced airflow at 4000 m MSL (REQ-ENV-004, reduced convective cooling), the PCB copper pour under/around the IC exposed pad must provide $\theta_{JA} \leq 40^\circ \text{C/W}$ to stay within the junction temperature budget at 45°C ambient. This shall be confirmed by thermal simulation and bench thermal imaging (VER-PWR-04).

6.4 PDN Target Impedance

Using the target impedance method (Ohm’s law applied to the worst-case transient step and allowable ripple budget):

$$Z_{target} = \frac{V_{DD} \times \text{ripple}\%}{\Delta I_{transient}}$$

For the 3.3 V_{DIG} rail, assuming a 3% ripple budget and a 300 mA transient step (SPI burst + IMU wake):

$$Z_{target,DIG} = \frac{3.3 \times 0.03}{0.3} = 0.33 \Omega$$

This target is straightforward to meet with standard MLCC decoupling (Table 12) and does not require an embedded-capacitance layer. The RF rail (3.3 V_{RF}) is instead governed by *noise floor* (PSRR of the TPS7A8300 at the switching frequency of the upstream buck and its harmonics) rather than target impedance alone, since the RTC6705 phase noise is directly modulated by supply noise.

6.5 Decoupling Strategy

Location	Value	Package	Purpose
STM32 VDD pins (each)	100 nF	0402	High-frequency local decoupling, <2 mm from pin
STM32 VDDA	100 nF + 1 μ F	0402/0603	Analog supply, star-connected, ferrite bead from VDD
BMI270 VDD/VDDIO	100 nF (each)	0402	Placed on the component side, direct via to GND plane
Buck output (bulk)	22 μ F $\times$ 2 (ceramic, X7R/X6S)	0805	Low-ESL bulk, close to inductor output
Buck output (bulk, electrolytic)	100 μ F polymer	SMD case D	Damps load-transient droop
Servo header (dedicated)	470 μ F electrolytic	Radial SMD	Absorbs stall-current spikes, see Section 6.6
RTC6705/PA rail	100 nF + 10 nF + 1 nF (staggered)	0402	Broadband decoupling across MHz–GHz range

Table 12: Decoupling capacitor placement and sizing strategy.

6.6 Servo Bulk Capacitance / Hold-Up Sizing

The buck converter’s control loop bandwidth is typically 1/10th to 1/5th of f_{sw} , giving an effective transient response time on the order of $t \approx 50 \mu\text{s}$ before the loop can react to a step load. Sizing the bulk capacitor to hold the rail within a 200 mV droop budget during a 1.5 A servo-stall current step:

$$C = \frac{I \cdot t}{\Delta V} = \frac{1.5 \times 50 \times 10^{-6}}{0.2} = 375 \mu\text{F}$$

The specified 470 μF electrolytic provides $\sim 25\%$ margin over this first-order estimate. **This capacitor must be physically located at the servo connector**, not solely at the buck output, since PCB trace/connector inductance ($\sim 1\text{--}5 \text{ nH/mm}$) will otherwise dominate the transient impedance seen by the servo.

6.7 RF Supply Isolation

The RTC6705 and RTC6659E PA **shall not** share a regulator, plane region, or return path with the servo or camera 5 V loads. The TPS7A8300 (PSRR typ. 60 dB at 1 MHz, derating at higher frequencies) is dedicated solely to the RTC6705 synthesizer core; the PA itself, being a saturated/non-linear stage, is comparatively tolerant of supply noise but shall still be fed through a ferrite-bead + local bulk/decoupling pi-filter from the main 5 V rail to prevent PA-generated switching noise (from RF envelope keying) from injecting back into the shared 5 V domain and desensitizing the camera’s analog CVBS output.

6.8 Input Protection

- **Reverse polarity:** P-channel MOSFET high-side protection (preferred over a series diode to avoid I^2R loss at 2.8 A) or an ideal-diode controller IC.
- **Overcurrent / short-circuit:** Input fuse or PTC resettable fuse sized at $1.5\times$ max steady-state current (~ 4.2 A rating for a 2.8 A load).
- **Inrush limiting:** NTC thermistor or MOSFET soft-start controller to limit inrush when hot-plugging a charged 6S pack into discharged input capacitance, preventing connector arcing/pitting.
- **Transient suppression:** TVS diode (e.g., SMBJ30A class, selected for standoff voltage above 25.2 V max pack voltage) across the input bus for motor/ESC-induced transients coupled back through the power distribution board.

7 PCB Stackup & Layer Definition

7.1 Stackup Table

A 6-layer stackup is mandatory to provide a continuous ground reference beneath both the high-speed SPI/digital signals and the 5.8 GHz RF trace, while allowing a dedicated power plane to keep PDN inductance low.

Layer	Function	Material	Thickness	Cu Wt.
L1	Signal (components, RF island)	—	—	1 oz (35 μm)
—	Prepreg	Isola 370HR, 2116, $\epsilon_r \approx 4.2$ @ 1 GHz	0.10 mm (4 mil)	—
L2	GND (solid reference plane)	—	—	1 oz
—	Core	Isola 370HR, $\epsilon_r \approx 4.2$	0.20 mm	—
L3	Power (5 V / 3.3 V split pour)	—	—	1 oz
—	Core	Isola 370HR, $\epsilon_r \approx 4.2$	0.71 mm (core, adjust for target thickness)	—
L4	Signal (SPI, UART, PWM)	—	—	1 oz
—	Prepreg	Isola 370HR, 2116	0.10 mm (4 mil)	—
L5	GND (solid reference plane)	—	—	1 oz
—	Core	Isola 370HR	0.20 mm	—
L6	Signal (bottom, low-speed / silkscreen side)	—	—	1 oz

Table 13: 6-layer stackup definition. Laminate specified as Isola 370HR (mid- T_g , well-characterized, cost-effective) for the digital stack. **Loss budget note:** the RF trace on L1 is short (<15 mm from RTC6659E to the U.FL connector); at this length, standard FR4-class loss (~ 0.35 – 0.45 dB/cm at 5.8 GHz) is acceptable (<0.7 dB total insertion loss) and does *not* justify the cost/lead-time penalty of a hybrid Rogers RO4350B cap layer for this program. This assumption shall be re-verified if the RF trace length grows beyond ~ 25 mm in final layout.

△ Caution — Fabricator Variation: Dielectric constant and thickness tolerances vary between fabricators and even between panel lots. The exact laminate opening (resin content, glass style) for each dielectric **shall be called out explicitly in the fab drawing** rather than left to the fabricator's standard stackup, and impedance-controlled layers shall be called out with a $\pm 10\%$ impedance tolerance requirement, verified by fabricator-supplied coupon TDR measurement per lot (Section 13).

7.2 Controlled Impedance Targets

Net Class	Topology	Target Z_0	Reference Layer
SPI (IMU, VTx, generic)	Single-ended microstrip	$50\ \Omega \pm 10\%$	L2 (solid GND)
UART (camera, GPS)	Single-ended microstrip	$50\ \Omega \pm 10\%$ (relaxed, low-speed)	L2
CVBS analog video	Single-ended microstrip	$75\ \Omega \pm 10\%$	L2
5.8 GHz RF (PA to antenna)	CPWG	$50\ \Omega \pm 5\%$	L2, plus coplanar ground pour on L1

Table 14: Impedance targets by net class. Note the CVBS video path is $75\ \Omega$ per composite video convention, distinct from the $50\ \Omega$ digital/RF targets – do not apply a blanket $50\ \Omega$ rule to the analog video trace.

8 Signal Integrity & Layout Rules

8.1 General Principles

Reflections arise from impedance discontinuities: changes in trace geometry, layer transitions without adjacent return-path stitching, connector transitions, and unterminated stubs. The single most effective SI control is maintaining a **solid, unbroken reference plane** directly beneath every high-speed or RF trace – this is non-negotiable for the SPI buses and the 5.8 GHz RF trace alike.

8.2 Digital Trace Rules (SPI / UART / PWM)

- Route SPI point-to-point (STM32 $\leftrightarrow$ BMI270, STM32 $\leftrightarrow$ RTC6705); no daisy-chaining, no stubs off the main line.
- Rule-of-thumb maximum unterminated trace length before transmission-line effects dominate: $\ell_{max} \approx t_r \cdot v_p / 6$, where t_r is signal rise time and v_p is propagation velocity ($\approx 1.5 \times 10^8$ m/s in FR4 microstrip). For a 170 MHz MCU with GPIO rise times on the order of 2–3 ns, $\ell_{max} \approx (2.5\text{ns} \times 1.5 \times 10^8) / 6 \approx 62\text{ mm}$ – comfortably longer than any on-board SPI run, so series termination is not strictly required but a 22–33 Ω series resistor at the driver is recommended as cheap insurance against ringing on the fastest (SPI to RTC6705) link.
- Maintain minimum $3\times$ trace-width spacing (“3W rule”) between adjacent parallel digital traces to limit crosstalk, increasing to 5W for the CVBS analog trace relative to any digital neighbor.
- Layer transitions (vias) on signal nets that change reference plane (e.g., L1 GND-referenced to L4 GND-referenced) do not by themselves introduce meaningful reflections at these edge rates; however, **a stitching via connecting the two reference planes shall be placed within $\sim 1.5\text{ mm}$ of every such signal via** to give the return current a short, low-inductance path – omitting the stitching via (not the signal via itself) is the actual common root cause of degraded signal quality at layer transitions.
- Right-angle (90°) corners are acceptable for all digital traces in this design at these edge rates and do not measurably increase EMI at these frequencies; 45° chamfers are used by convention for manufacturability/etch consistency rather than as an SI requirement.

8.3 RF Trace Design (5.8 GHz, RTC6659E to U.FL)

The PA-to-antenna trace **shall** be implemented as a $50\ \Omega$ Coplanar Waveguide with Ground (CPWG), referenced to the L2 ground plane below and flanked by a stitched coplanar ground pour on L1.

Design Note — Illustrative CPWG Solution (verify with 2D field solver): For the L1-to-L2 dielectric height of $h = 0.10\text{ mm}$ and $\epsilon_r \approx 4.2$ specified in Table 13, a representative solver-derived

solution achieving $\approx 50\ \Omega \pm 5\%$ is: trace width $W \approx 0.22\text{--}0.25\text{ mm}$, coplanar gap $G \approx 0.13\text{--}0.16\text{ mm}$ to the adjacent ground pour, 1 oz copper. **These values are illustrative only** and shall be confirmed with a 2D field solver (e.g., Polar SI9000, Saturn PCB Toolkit, or the fabricator’s stackup calculator) against the as-built dielectric thickness and resin content before tape-out, per REQ-SI-002 below.

8.3.1 Ground Stitching Via Fence

To suppress parallel-plate/coplanar mode leakage and maintain waveguide behavior, a via fence of grounded stitching vias shall flank the CPWG trace on both sides. Guided wavelength at 5.8 GHz for $\varepsilon_{r,eff} \approx 3.0$ (typical effective permittivity for a CPWG on FR4-class material, accounting for field lines partly in air):

$$\lambda_g = \frac{c}{f\sqrt{\varepsilon_{r,eff}}} = \frac{3 \times 10^8}{5.8 \times 10^9 \times \sqrt{3.0}} \approx 29.9\text{ mm}$$

Applying the standard $\lambda_g/10$ to $\lambda_g/20$ via-spacing guideline yields a maximum stitching via pitch of **1.5–3.0 mm**; this design specifies **$\leq 2.0\text{ mm}$** pitch for margin, with via-to-trace-edge spacing kept as small as the design rules allow ($\geq 0.2\text{ mm}$ clearance to the coplanar ground pour edge).

8.4 Vias, Corners, and Stubs – Consolidated Rules

Feature	Rule
Signal vias (digital)	Permitted freely; pair with a ground stitching via for reference-plane continuity (Section 8, digital rules).
Signal vias (RF, 5.8 GHz)	Minimize; if unavoidable (e.g., L1-to-L3 transition), use back-drilling or a blind via to remove the unused stub capacitance, and surround with a ground via fence at $\leq 2\text{ mm}$ pitch.
Right-angle bends (digital)	Acceptable; no measurable SI or EMI penalty at these edge rates.
Right-angle bends (RF)	Not acceptable – use smooth arcs or 45° /mitered bends for the 5.8 GHz CPWG; sharp corners create a local impedance discontinuity and radiating discontinuity that is significant relative to the guided wavelength at microwave frequencies.
Stubs (all nets)	Prohibited. Any unused via stub, test point spur, or dangling net segment on a controlled-impedance or RF net shall be removed or back-drilled.

8.5 Signal Integrity Verification Requirements

- **REQ-SI-001**: All controlled-impedance nets shall be called out on the fab drawing with target impedance and tolerance per Table 13/impedance targets table.
- **REQ-SI-002**: RF trace geometry shall be confirmed by field-solver simulation prior to layout freeze, and by TDR measurement on a fabricated coupon before production release.
- **REQ-SI-003**: A minimum of one ground stitching via per 1.5 mm shall be placed adjacent to every signal via that changes ground reference plane on SPI/UART nets.

9 RF Subsystem Design (VTx)

9.1 RTC6705 Configuration

The RTC6705 integrates the PLL frequency synthesizer and FM video modulator. The MCU configures the output frequency and power level over a 3-wire SPI-like interface by writing to internal

25-bit registers (frequency synthesis register N/A/R divider values per the RTC6705 datasheet register map). Firmware shall abstract this behind a simple `vtx_set_channel(band, channel)` / `vtx_set_power(level)` API, with the resolved register values validated in unit test against the vendor reference table.

9.2 Standard 5.8 GHz Band Plan

The firmware shall support, at minimum, the following industry-standard band plan (channel numbering and frequencies consistent with common FPV goggle/VRx convention):

Band	CH1	CH2	CH3	CH4	CH5	CH6	CH7	CH8
A (Boscam A)	5865	5845	5825	5805	5785	5765	5745	5725
B (Boscam B)	5733	5752	5771	5790	5809	5828	5847	5866
E (Boscam E / DJI)	5705	5685	5665	5645	5885	5905	5925	5945
F (Fatshark / IRC)	5740	5760	5780	5800	5820	5840	5860	5880
R (Raceband)	5658	5695	5732	5769	5806	5843	5880	5917

Table 16: 5.8 GHz analog FPV band plan, frequencies in MHz. All 40 channels shall be exposed in firmware; region-locked builds shall mask out-of-band channels per REQ-EMC-003 at compile time or via a region-configuration record in non-volatile storage, not left to the end user to self-police.

⚠ Caution — Regulatory Scope: Bands outside the 5.725–5.875 GHz ISM allocation for a given region (e.g., portions of Band E) may fall outside the legal FCC/ETSI allocation depending on jurisdiction. Firmware region-lock tables shall be maintained and reviewed against current regulatory publications at each firmware release; this document does not substitute for a regulatory compliance sign-off.

9.3 Power Level Control

Three discrete power levels shall be supported, mapped to RTC6659E PA bias/gain control:

Level	Nom. Power	Typ. Use Case	Regulatory Fit
Low	25 mW	Default/boot state, close-range bench test	EU/ETSI EN 300 440 compliant
Medium	200 mW	Regional (e.g., select non-EU jurisdictions)	Verify local allocation before use
High	400 mW	Long-range observation missions	US FCC Part 15.247 typical allowance (verify EIRP with antenna gain)

Table 17: VTx power levels. Firmware shall boot to Low by default (REQ-EMC-003) and require an explicit, logged configuration change to enable Medium/High.

9.4 Thermal Management of the PA

The RTC6659E dissipates the difference between DC input power and RF output power as heat (efficiency on the order of 30–40% typical for this PA class at 400 mW output, implying ~0.6–0.9 W dissipated as heat at full power). A thermal via array (≥ 9 vias, 0.3 mm drill, on 1.0 mm pitch) beneath the PA’s exposed thermal pad shall connect to an internal copper pour on L3/L5 to spread heat, and firmware shall implement the PA thermal throttle per REQ-SAF-003.

10 EMI/EMC & Shielding Strategy

10.1 Physical Partitioning

- The RTC6705/RTC6659E RF island shall be located at the board edge nearest the antenna connector, and separated from the BMI270 IMU and the compass by a minimum of **25 mm** or by an intervening ground-via “moat,” whichever provides greater practical isolation.
- The compass (QMC5883L) shall additionally be kept clear of high-current DC paths (buck inductor, servo header, battery input traces) by a minimum of **15 mm**, since DC bias fields from switching currents directly corrupt magnetometer readings, independent of RF concerns.
- The analog CVBS trace shall be routed away from the RF island and from the buck converter’s switching node, both of which are broadband noise sources capable of coupling into the unshielded composite video signal and appearing as visible interference (“rainbow noise”) in the downlink image.

10.2 RF Shielding Can

A grounded, tin-plated (or nickel-silver) RF shielding can footprint shall be provided over the RTC6705 + RTC6659E + matching network area, soldered to a continuous ground-via fence around its perimeter (pitch ≤ 2 mm, consistent with Section 8) to present a low-impedance return path for the can itself. The can shall include a cutout or feedthrough for the CPWG trace to the antenna connector, with the gap treated as a slot antenna risk and minimized (< 0.5 mm clearance beyond the trace/coplanar ground).

10.3 Conducted/Coupled Noise Mitigation

- Ferrite bead + local decoupling “pi filter” on the PA’s 5 V feed (Section 6) to block PA envelope-keying noise from the shared 5 V domain.
- Common-mode choke on the UART lines to the camera module and GPS module where cable runs exceed ~ 50 mm, to suppress common-mode RF pickup on these longer runs.
- Ground stitching vias (population ≥ 1 per 100 mm^2) across the full board to minimize ground-plane impedance and suppress ground-bounce coupling between the digital and RF/analog domains.

10.4 Pre-Compliance Test Plan

- Near-field H-field/E-field probe scan across the board with a spectrum analyzer to identify unintended radiators prior to formal compliance testing.
- Conducted emissions on the battery input leads (relevant if this FC is later integrated with a shared power bus alongside other RF-sensitive payloads).
- IMU/compass noise-floor comparison with VTx PA keyed on vs. off (REQ-EMC-002 verification).
- Full FCC Part 15.247 / ETSI EN 300 440 formal compliance testing at an accredited lab prior to any commercial release (out of scope for internal pre-compliance testing described here).

11 DFM / Mechanical Reliability

11.1 Mounting & Vibration Isolation

The board shall use the industry-standard **30.5 mm × 30.5 mm** FC mounting pattern with **4 mm diameter** mounting holes to accept silicone/rubber vibration-damping grommets. Grommet durometer shall be selected (typically 55A–65A Shore hardness) to filter motor/propeller-induced vibration frequencies (typically 80–400 Hz fundamental for the target propeller sizes) from reaching the BMI270

IMU, consistent with REQ-ENV-002. Grommet compression at installation shall be specified at 10–15% of free height to maintain consistent damping without bottoming out under sustained 1 g load plus vibration.

11.2 Board Outline & Connector Keepouts

- Maintain a minimum 3 mm keepout from board edge to any component taller than 2 mm, to accommodate a protective frame/standoff without interference.
- U.FL antenna connector placed at the board edge with the RF trace as short and direct as practical (Section 8).
- Power input (XT30/XT60 pigtail or bullet-connector footprint) shall be routed with ≥ 1.5 mm trace width per ounce of copper per amp of continuous current (approx. 20 mil/A at 1 oz copper, 10°C rise) for the 2.8 A transient budget – i.e., a minimum 2.0 mm (80 mil) power trace width on 2 oz copper for the main input, or equivalent via-stitched multi-layer pour.

11.3 Fabrication & Assembly Class

- Fabrication per IPC-6012 Class 2 (general electronics) as the baseline; Class 3 workmanship notes applied specifically to the RF island solder joints and the servo/power connectors, which see the highest mechanical and thermal cyclic stress.
- Assembly (solder joint) acceptance per IPC-A-610 Class 2.
- Minimum annular ring, minimum drill size, and minimum trace/space shall be confirmed against the selected fabricator’s standard capability table at DFM review, with ≥ 4 mil/4 mil (0.1 mm/0.1 mm) trace/space assumed as the baseline design rule for this stackup.

11.4 Conformal Coating

Per REQ-ENV-003, a conformal coating (acrylic or silicone-based, selected for rework compatibility) shall be applied post-assembly, with the U.FL connector, SD card slot, and any user-adjustable connectors masked prior to coating. Coating shall not be applied over the RF shielding can’s ground contact fingers if a spring-finger can is used, to preserve electrical continuity.

11.5 Connector Selection

Interface	Connector	Notes
Battery input	XT30 (pigtail) or solder pads	Rated ≥ 30 A continuous; oversized for this load, standard for the platform class
Camera video/control	JST-SH 1.0 mm, 4-pin	Matches RunCam Split 4 V3 factory pigtail
Servo	JST-SH 1.0 mm, 3-pin (or bare pads)	GND/5 V/PWM per Section 1
Antenna	U.FL (or MMCX for field durability)	U.FL has limited mating cycle life (~ 30); MMCX recommended if frequent antenna swaps expected in the field

Table 18: Connector selection summary.

12 Firmware & Control Architecture

12.1 Task Scheduling

A bare-metal, interrupt-driven scheduler (rather than a full RTOS) is recommended given the bounded, well-characterized task set, consistent with common flight-controller firmware architecture:

Task	Rate	Priority
IMU sampling (BMI270, SPI DMA)	6.4–8 kHz	Highest (hardware timer + DMA complete IRQ)
Attitude/rate control loop	2–4 kHz	High
Barometer fusion	50 Hz	Medium
GPS parse (UART3 + PPS capture)	10 Hz (nav), async PPS	Medium
Servo PWM update (double-buffered)	50 Hz	Low
Camera UART housekeeping / telemetry	10 Hz poll, event-driven commands	Low
VTx SPI configuration	Event-driven (channel/power change only)	Low
Battery/thermal monitoring (ADC)	10 Hz	Medium (safety-relevant)

Table 19: Firmware task rates. The control loop (2–4 kHz) exceeds REQ-PERF-002’s 2 kHz minimum with margin for future gain-scheduling or notch-filter additions.

12.2 Camera Control Protocol

The UART link to the RunCam Split 4 V3 shall implement the RunCam Device Protocol (or equivalent vendor-published serial protocol) to: (a) start/stop SD recording, (b) query recording/SD status for OSD display, and (c) optionally forward OSD text elements for overlay on the CVBS output. A 3-retry/200 ms-timeout handshake (Section 4.3) governs link-health detection.

12.3 Fault Handling & Watchdogs

- Independent hardware watchdog (IWDG) on the STM32, refreshed only from the main control loop context, to force a reset on control-loop lockup.
- Brown-out reset (BOR) threshold configured above the minimum operating voltage of the 3.3 V_{DIG} rail with margin, to guarantee clean reset behavior rather than erratic operation during a sagging bus.
- VTx channel/power configuration persisted to internal flash (or EEPROM emulation) so a power cycle resumes on the last configured channel without requiring re-pairing with the ground receiver.
- GPS fix loss: position-hold mode shall gracefully degrade to altitude-hold-only (barometer) with a telemetry warning, rather than an uncommanded mode fault.

13 Verification & Validation Plan

Req. ID	Test Method	Acceptance Criteria
REQ-CAM-001/002	Bench recording test, 30-min continuous 4K@30 + simultaneous CVBS capture	No dropped frames (verified via frame-hash continuity check); CVBS signal within CCIR/EIA timing spec on waveform monitor
REQ-GIM-001/002	Bench servo sweep + RC failsafe injection	Full 120° mechanical travel confirmed; failsafe-to-safe-position time ≤ 500 ms measured via oscilloscope on PWM line and optical encoder on servo horn
REQ-VTX-001/002	SPI register readback + spectrum analyzer channel/power verification	Measured frequency within ± 1 MHz of Table 16; measured EIRP within ± 1 dB of configured level

Req. ID	Test Method	Acceptance Criteria
REQ-PERF-001	Glass-to-glass latency test (high-speed camera comparing live scene to goggle display)	≤ 80 ms measured
REQ-PERF-002	IMU/loop-rate logging via debug UART	Confirmed ≥ 6.4 kHz IMU, ≥ 2 kHz control loop in flight log
REQ-PERF-003	Range test, line-of-sight, calibrated attenuator bench pre-test then field range test	≥ 2 km achieved with ≥ 6 dB link margin
REQ-ENV-002	MIL-STD-810H Method 514.8 random vibration, 3-axis	No solder joint failure ($5\times$ magnification inspection), no component detachment, IMU output noise floor within spec post-test
REQ-EMC-002	IMU/GPS/baro noise floor, PA keyed vs. unkeyed	≤ 1 dB (or 1 LSB-equivalent) degradation
VER-PWR-03	Simulated power-loss during active SD write (bench PSU step to 0 V)	File system closes cleanly, no corrupted video container, confirmed across 20 repeated trials
VER-PWR-04	Thermal imaging of buck/PA under sustained max load, 45°C chamber, no airflow	Junction/case temperatures within datasheet limits with $\geq 15^{\circ}\text{C}$ margin
REQ-SI-002	TDR measurement on fabricated coupon	CPWG impedance $50\ \Omega \pm 5\%$; SPI/UART microstrip $50\ \Omega \pm 10\%$

Table 20: Verification cross-reference matrix (representative subset; full matrix maintained in the program’s requirements-management tool).

14 Risk Register / FMEA (Summary)

Component/ Function	Failure Mode	Effect	Sev.	Mitigation
RTC6659E PA	Thermal runaway under sustained 400 mW	Video downlink loss, potential PA damage	High	Thermal via array + firmware thermal throttle (REQ-SAF-003)
Servo	Mechanical stall / winding burnout	Loss of camera aim, potential rail brown-out	Med	Current monitoring + auto-neutral (Section 4.3)
SD card	Card failure / write corruption mid-flight	Loss of recorded footage (non-flight-critical)	Med	Bulk capacitance hold-up (Section 6.6); status telemetry to pilot

Component/ Function	Failure Mode	Effect	Sev.	Mitigation
Compass	Magnetic interference from nearby current loops	Heading drift, navigation error	High	Physical separation rules (Section 10); in-flight compass health monitor comparing GPS course vs. heading
Buck converter	Output overvoltage (component failure, latch-up)	Overvoltage to 5 V loads, potential camera/servo damage	High	Consider adding an output crowbar/OVP clamp – tracked as OPEN-04
VTx region-lock table	Firmware misconfiguration ships with wrong region default	Regulatory compliance	non-High	Release build process shall assert region default at compile time with a mandatory build-config check

Table 21: Summary risk register. Full FMEA with occurrence/detection scoring (RPN) maintained in the program risk-tracking system.

15 Open Items / Action Log

ID	Description	Owner
OPEN-01	Confirm airframe housing provides ≥ 3 mm ventilation clearance around the camera processing board to avoid sustained-recording thermal throttling.	Mech. Eng.
OPEN-02	Select and validate the digital 3.3 V LDO part (placeholder TLV75533) against noise/PSRR requirements for the MCU/IMU rail.	Elec. Eng.
OPEN-03	Evaluate second-source RF chipset (e.g., IRC Tramp-compatible) against RTC6705/RTC6659E for supply-chain risk mitigation.	Procurement / RF Eng.
OPEN-04	Evaluate adding output overvoltage protection (crowbar or clamp) on the 5 V rail per FMEA risk item.	Elec. Eng.
OPEN-05	Confirm final regulatory target markets to lock the default/maximum VTx power level and region-lock table contents (originally raised as: <i>which 5.8 GHz bands – FatShark, Raceband, etc. – must firmware support?</i>).	Program Mgmt.
OPEN-06	Confirm connector choice (U.FL vs. MMCX) based on expected field antenna-swap frequency.	Mech. Eng.

ID	Description	Owner
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Table 22: Open items requiring resolution before Critical Design Review (CDR).

A Appendix A: STM32G473CEU6 Pin/Peripheral Mapping (Preliminary)

Pin	Function	Peripheral	Net
PA5/PA6/PA7	SPI1 SCK/MISO/MOSI	SPI1	IMU_SCK/MISO/MOSI
PA4	SPI1 CS	GPIO	IMU_CS
PB13/PB14/PB15	SPI2 SCK/MISO/MOSI	SPI2	VTX_SCK/MISO/MOSI
PB12	SPI2 CS	GPIO	VTX_CS
PB6/PB7	I2C1 SCL/SDA	I2C1	BARO/COMPASS bus
PC10/PC11	UART3 TX/RX	USART3	GPS TX/RX
PB10/PB11	UART3 alt. or UART4	USART	Camera control UART
PA0 (TIM2_CH1)	Servo PWM	TIM2	GIMBAL_PWM
PA1	PPS capture input	TIM input capture	GPS_PPS
PC0 (ADC1_IN6)	Battery voltage sense	ADC1	VBAT_SENSE
PC1 (ADC1_IN7)	Servo current sense	ADC1	SERVO_ISENSE

Table 23: Preliminary pin mapping. Final assignment subject to PCB layout/routing optimization; this table shall be kept in sync with the schematic net list at each revision.

B Appendix B: References

1. IPC-2221, IPC-2141A, IPC-6012, IPC-A-610 (IPC International, Inc.)
2. STM32G473xx Datasheet and Reference Manual (RM0440), STMicroelectronics
3. BMI270 Datasheet, Bosch Sensortec
4. DPS310 Datasheet, Infineon Technologies
5. MAX-M10S Integration Manual, u-blox
6. RTC6705 / RTC6659E Datasheets, RichWave Technology
7. TPS54302 / TPS7A8300 Datasheets, Texas Instruments
8. FCC Part 15, Subpart C; ETSI EN 300 440 (regulatory text, respective publishing bodies)
9. MIL-STD-810H, Method 514.8 (Vibration), U.S. Department of Defense
10. Bogatin, E., *Signal and Power Integrity – Simplified*, for general PDN target-impedance and via/right-angle-bend engineering guidance referenced qualitatively in Section 8.